

Tutorial 1 – Getting Started with MicroPython

Aim

MicroPython is a lean and efficient implementation of the Python 3 programming language that includes a small subset of the Python standard library and is optimised to run on microcontrollers and in constrained environments. This tutorial will guide you through the installation of development environment and show you how to write and run your first MicroPython program on the Arduino Nano 33 BLE Sense Rev2.

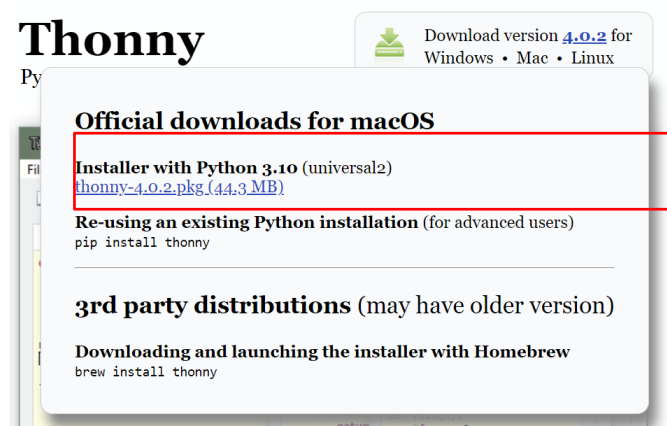
Thonny IDE

Thonny is a Python IDE that also supports MicroPython development on Windows, Linux and MacOS. We recommend you download the latest version from the official website (<https://thonny.org/>) for your platform.

For Windows users, please download the latest 64-bit installer:



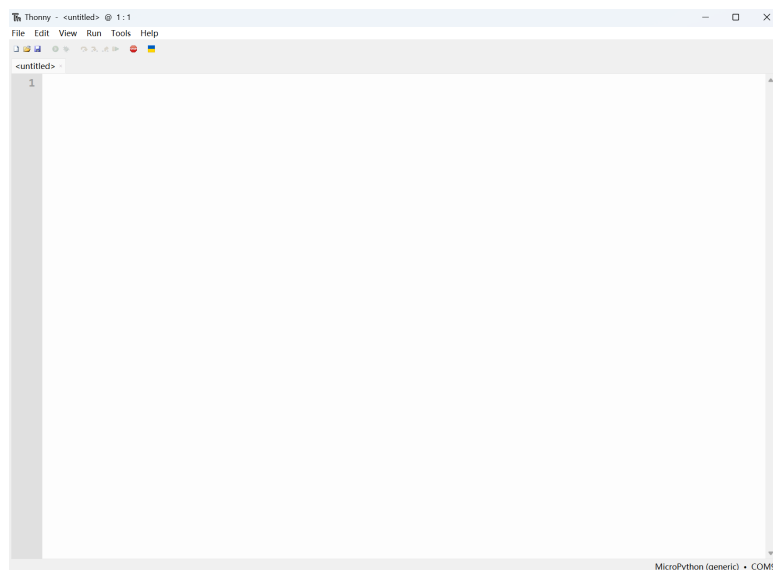
For MacOS users, please download latest installer:



For Linux users, please use the bash command to install the latest version:



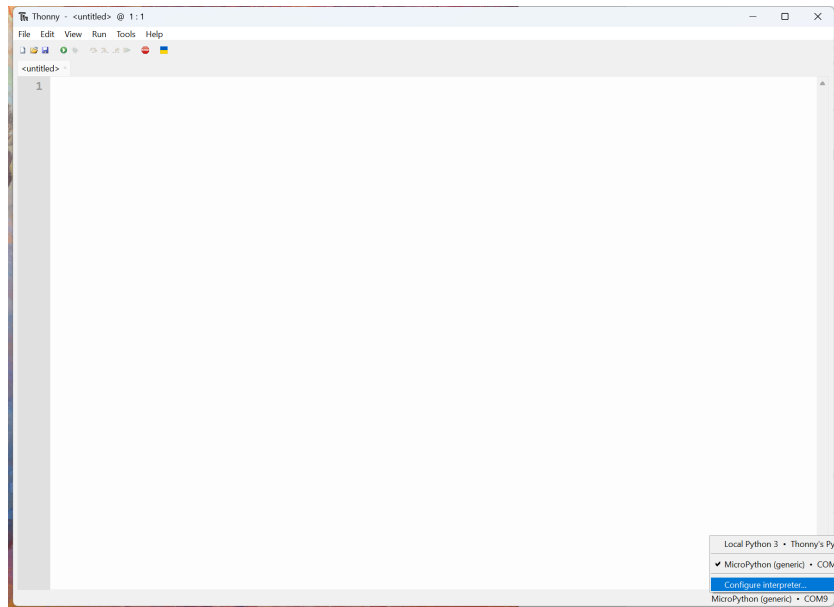
Open the Thonny IDE once it is installed, you will see the following text editor view:



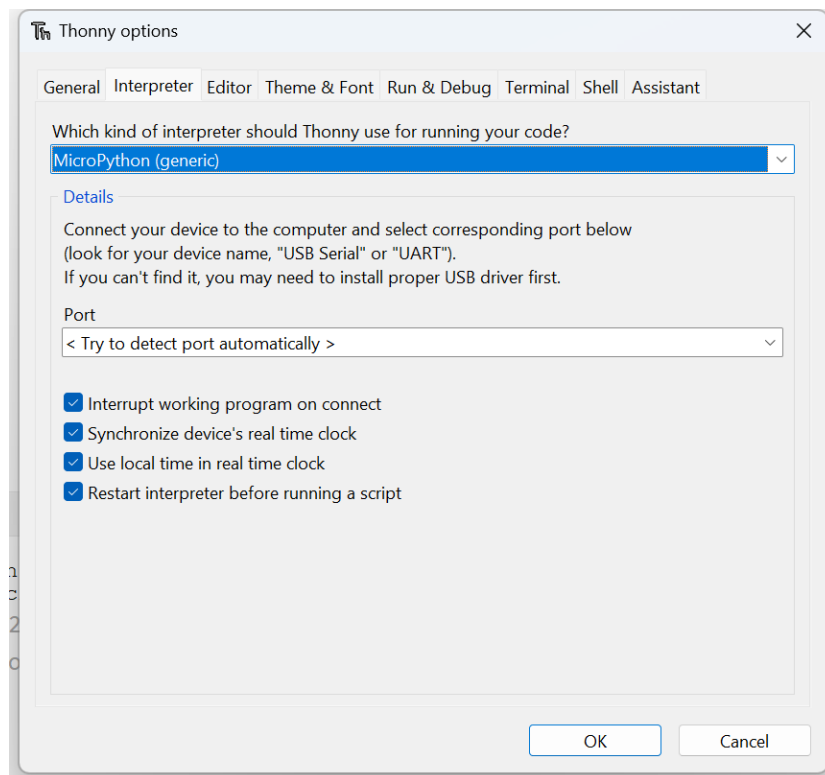
Hello MicroPython

Connect the Arduino Nano 33 BLE Sense board to your computer. Please check it corresponding port via device manager (Windows) or “ls /dev/tty*” on Linux and MacOS.

Open the Thonny IDE, click “Configure interpreter”:

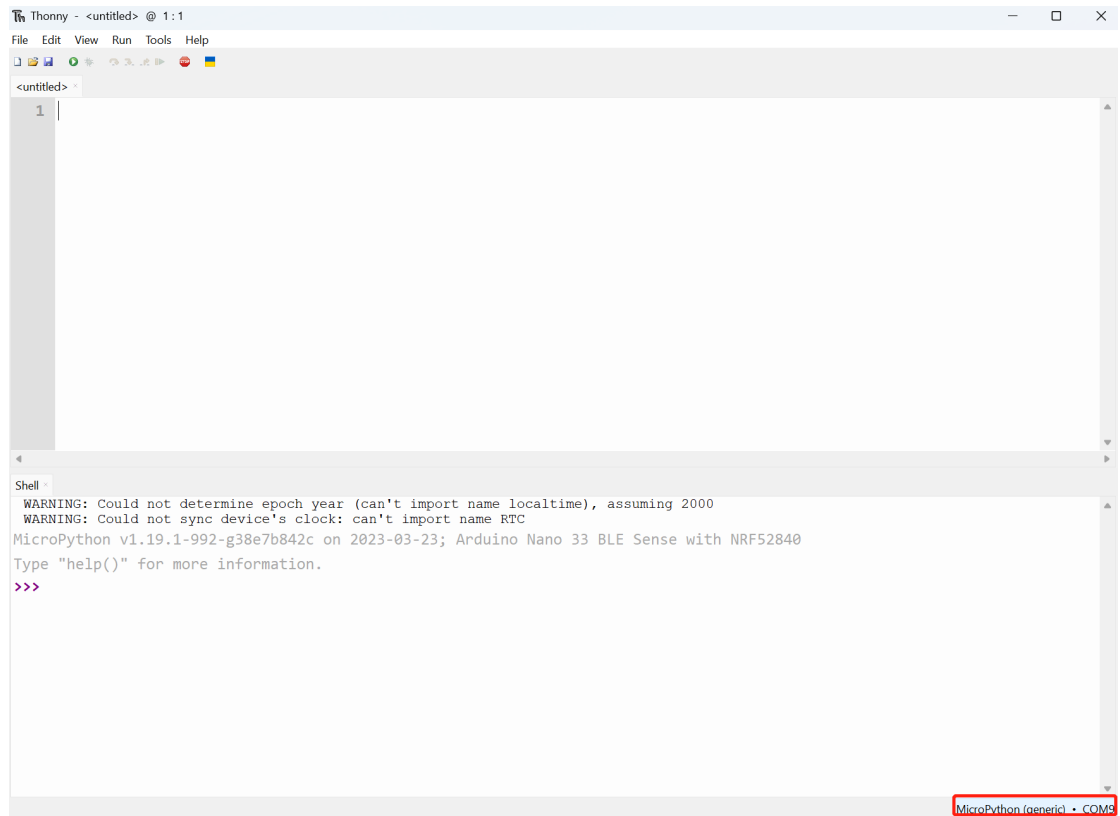


You will see the pop-up window:

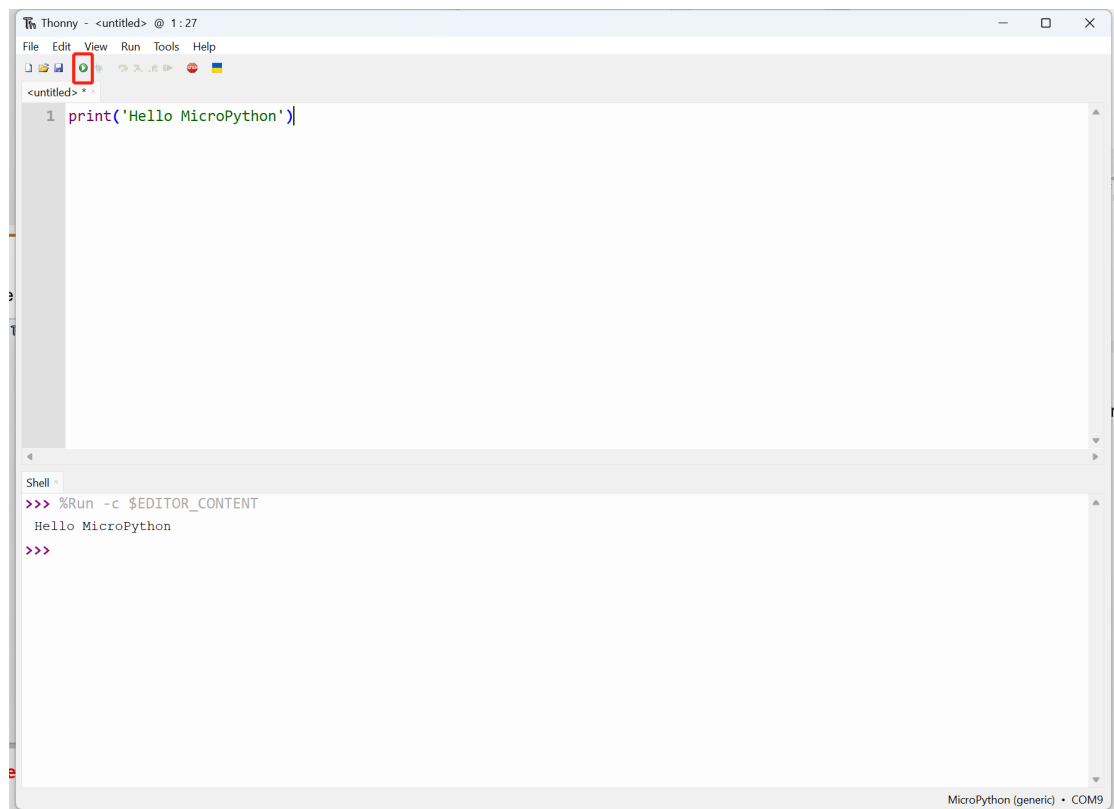


Please select MicroPython (generic) as the interpreter and the correct port number

After configuring the interpreter, you should be able to connect the Arduino board:



Now, write your first MicroPython Script in the text editor and press the run (green) button:



You can see that the “Hello MicroPython” has been printed in the MicroPython’s REPL prompt. In the following tutorials or labs, you can print/read sensor data in this way.

Press the save button or CTRL+S to **save the script on your computer:**



Please do not save your scripts on the MicroPython Device unless necessary (project demo). This will keep the longevity of the flash memory.

Conclusion

This tutorial has introduced the installation of the Thonny IDE and the MicroPython development on it.